

4.1

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = z \quad \text{tak, z.e.} \quad z = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

optimalizace:

$$A^T \cdot A \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = A^T z$$

$$A^T = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

tedy hledáme

 $A^T \cdot A$

$$A^T \cdot A = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T \cdot z = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

dosadíme

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2/5 \\ 0 & 1 & 0 & 4/5 \\ 0 & 0 & 1 & 3/5 \end{array} \right)$$

tedy

$$a = \frac{2}{5}$$

$$b = \frac{4}{5}$$

$$c = \frac{3}{5}$$